Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/24/2007
PI Name:	Muller, Susan J Ph.D.		
Project Title:	A Biosensor For Single-Molecule DNA Sequencing		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Biofluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1462	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:	Ph.D. students Victor A. Beck and Nerayo P. Teclemariam have both completed their doctoral degrees, as have the undergraduate researchers associated with this project.		
COI Name (Institution):	Shaqfeh, Eric (Department of Chemical Engineering, Stanford University)		
Grant/Contract No.:	NNC04GA72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this project is to develop a biosensor for rapid, single-molecule DNA screening for specific genes which may be indicators of susceptibility to space radiation damage. The basic concept is to use hydrodynamics to fully extend genomic DNA in a microfluidic device. Fluorescent beads with surface-bound, complementary oligonucleotide probe sequences (corresponding to genes or sequences of interest) will be introduced and hybridized to the stretched DNA. Any unhybridized beads will be separated from the DNA-bead complexes. Imaging of the bead positions along the stretched, linear DNA backbone will then provide sequence information since the beads act as position and sequence sensitive markers. This project is a collaboration involving coordinated experimental studies at the University of California, Berkeley and simulations at Stanford University. To realize the concept as described in the Task Abstract, we have identified six specific objectives: 1) Identify flow geometries (converging flows, post arrays, crossed slots) and the flow driving force (pressure, electro-osmostic, electrophoretic) that will be most effective in stretching DNA in a microfluidic device. 2) Determine the rate at which the beads sample the extended DNA and the binding kinetics for formation of the complementary DNA-bead complex; bead transport issues will establish the optimal bead size. 3) Quantify transport of the DNA-bead complex through the microdevice and its relaxation dynamics. This will determine the time scale available for imaging. 4) Examine the sieving and separation of unbound beads from the DNA-bead complex. As envisaged in our test design, this will be in a single pass through a post array. 5) Determine the optimal means of analysis of the DNA-bead complex now separated from the remaining unbound beads. 6) Determine the resolution and limits(e.g., size of DNA, chain scission constraints) of the use of this technique for sequencing. This research is expected to be valuable to NASA for screening for genetic damage from the highly-charged, energetic heavy ions (HZE particles) which are the biologically dangerous component of space radiation. The potential of rapidly screening single DNA molecules for specific genes will aid in understanding genetic sensitivity and resistance to damage by space radiation. Ultimately, this microfluidic-based approach may provide an efficient method for determining an individual's susceptibility to space radiation-induced cancers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The biosensor would be valuable for genetic screening and sequencing in a host of applications. Moreover, we anticipate that the studies of how to manipulate large molecules using hydrodynamics will affect a broad range of bioMEMS applications, genomics, and DNA nanotechnology. Of particular interest are 1) the long read lengths potentially available (of order 10's of Mbps) and 2) the possibility of recovering single pieces of DNA of particular interest for subsequent analysis or amplification.
Task Progress:	The ultimate goal of this project is to develop a biosensor for rapid, single-molecule DNA screening for specific genes which may be indicators of susceptibility to space radiation damage. The basic concept is to use hydrodynamics to fully extend genomic DNA in a microfluidic device. Fluorescent beads with surface-bound, complementary oligonucleotide probe sequences (corresponding to genes or sequences of interest) will be introduced and hybridized to the stretched DNA. Any unhybridized beads will be separated from the DNA-bead complexes. Imaging of the bead positions along the stretched, linear DNA backbone will then provide sequence information since the beads act as position and sequence sensitive markers. This project is a collaboration involving coordinated experiments at Berkeley and computer simulations at Stanford. Work has focused on engineering the first step, namely, devising a microfluidic flow geometry to stretch out long DNA chains to near full extension. Accomplishments include: 1) evaluating flow geometries based on arrays of obstacles as a way to stretch DNA, 2) identifying the factors that control the stretching of individual DNA molecules, 3) optimizing these array geometries for DNA stretching, 4) validating simulation tools we have developed through a detailed comparison of direct visualization experiments and Brownian simulations of DNA dynamics in these complex flows, and 5) identifying new geometries and strategies for stretching DNA for genotyping. Guided in part by the literature on the electrophoresis of DNA through gels, we have studied a range of flow geometries based on arrays of posts in a microchannel. As in gel electrophoresis, we anticipate that large DNA molecules will interact with the obstacles (here, the posts; in gel electrophoresis, the gel fibers), hook around the obstacles, and stretch through a rope-over-a-pulley release mechanism. We find that flow strength (the ratio of convection to diffusion), post size (relative to the DNA size), the initial position of a DNA molecule with respect to the nearest post, and array geometry control the degree of chain stretching. Array geometry is critical because head on collisions with posts are much more effective than glancing ones, and any open channels allow some fraction of the DNA molecules to exit the array without undergoing any collisions or hooking interactions. Relaxation of the DNA conformation within the array following a hook is also important, and long, extended, regular arrays are not easily optimized for stretching of DNA. Based on these considerations, three different regular post array geometries were designed and tested via extensive direct visualization experiments using devices microfabricated in silicon and fluorescence microscopy to track and image DNA molecules labeled with a fluorescent dye. High aspect ratio posts of diameters of order 1 micron were created via deep reactive ion etching, this produced posts with diameters comparable to the radius of gyration of the lambda-bacteriophage DNA used in these studies. The high aspect ratio of the posts allowed us to treat the flow field as essentially two-dimensional. The initial position and extension of DNA molecules just upstream of the array were measured and used as inputs for corresponding Brownian dynamics simulations over a range of flow strengths. DNA molecular extensions and positions were then measured as a function of position within the array for large ensembles of molecules and at varying flow strengths, and compared to simulation results. Inspection of both the experimental and simulation movies shows the complex motion of the DNA molecules through the array. As noted by other researchers, a cyclical pattern of DNA motion is present which can be broken down into three distinct parts i) DNA collision with a post and hook formation, ii) extension in the flow field and release with a rope-over-pulley or extended mechanism, and iii) relaxation towards its equilibrium configuration and collision with another post at some downstream distance. A more detailed comparison shows excellent agreement between experiments and simulations was obtained at low flow strengths. The fraction of stretched molecules at varying position within the array, the position at which the molecules undergo first and second hooks, and the distribution of stretched molecules as a function of starting position normal to the flow were all faithfully reproduced by Brownian dynamics simulations. We find that relatively few columns of posts are needed in the optimized array, since most chains hook within the first several columns. Relaxation of the stretched

	DNA conformations within the array means that longer arrays are not necessarily more effective than shorter arrays. At higher flow strengths, the agreement between experiments and simulations is somewhat poorer, possibly due to effects of ensemble size; at higher flow strength, the number of molecules needed for the simulations to converge becomes larger than can be accessed in the experiments. However, the agreement between the experiments and simulations with respect to hooking location suggests that the velocity field, initial offset parameter, and diffusivity of the DNA are sufficient to at least qualitatively describe the hooking dynamics via simulations. Moreover, the qualitative and quantitative comparisons of experiments and Brownian dynamics simulations validate the simulation tool, which can be used to guide future designs. In addition, a scaling argument developed as part of this work appears to work well in guiding array design: essentially, the spacing between posts normal to the flow direction should be chosen such that a molecule that “misses” the first column of posts that it encounters will have time to diffuse normal to the flow the distance required for it to collide with a post in the next column it encounters. Ultimately, however, we believe that stagnation point flows may be the best route to single molecule genotyping and analysis. A prototypical stagnation point is generated at the intersection of two channels that cross at 90 degrees when fluid flows in through two opposed channels and out through opposed channels at 90 degrees to the inlets. In such a flow, a DNA molecule may be trapped for minutes or hours at a stationary stagnation point, stretched to the desired fractional extension by controlling the flow strength, examined for the presence or absence of particular target sequences, and then sorted into one or the other outlet channels by controlling the pressure at the exit reservoirs. We are in the process of designing and testing a range of stagnation point flows for hybridization studies, for genotyping, and for studies of single molecule reactions.
Bibliography Type:	Description: (Last Updated: 12/26/2007)
Articles in Peer-reviewed Journals	Teclemariam NP, Beck VA, Shaqfeh ESG, Muller SJ. "Dynamics of DNA polymers in post arrays: Comparison of single molecule experiments and simulations." Macromolecules. 2007 May 15; 40(10): 3848-59. http://dx.doi.org/10.1021/ma062892e , May-2007
Articles in Peer-reviewed Journals	Teclemariam NP, Muller SJ. "DNA transport between hooks in microfabricated post arrays." Macromolecules, in preparation, December 2007. , Dec-2007
Dissertations and Theses	Beck VA. "Polymer dynamics in nonlinear and microfluidic flows." Dissertation, Stanford University, September 2006. , Sep-2006
Dissertations and Theses	Teclemariam NP. "Controlling dynamics and conformation of DNA in flow through microfabricated post arrays: A single molecule study." Dissertation, University of California, Berkeley, December 2007. , Dec-2007